

1	Course Name:
	INTELLIGENT DIGITAL TWIN
2	Course Code:
	IQB20803
3	Year and Semester Offered:
	Year 2 Semester 3
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course introduces students to the theoretical perspective and concept of digital twins, highlighting the importance of adopting this technology across various industries. It examines the environmental, social, and governance impacts of utilizing digital twin technology. Additionally, the course allows students to explore the ethics, security, and privacy concerns related to digital twins. As a result, students will gain an understanding of future trends and challenges associated with digital twin technology.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Apply the theoretical aspects of digital twin technology. 2. Assess the impact of adopting digital twin technology for environmental, social and governance. 3. Evaluate the usage of digital twin technology in education and industry.
8	Content Outline:
	Topic
	1. Theoretical foundations of digital twin technology
	2. Concept of digital twin
	3. Digital twin architecture
	4. Digital twin market and industries
	5. Sustainability and environmental impact
	6. Ethics, security, privacy, and governance in digital twin
	7. Future trends and challenges in digital twin
9	References:
	Chaudhary, Gopal, Khari, Manju, Elhoseny, and Mohamed (2024). Digital Twin Technology. CRC Press (GB), ISBN9780367677978.